



Pseudo-boiling and heat transfer deterioration while heating supercritical liquid rocket engine propellants

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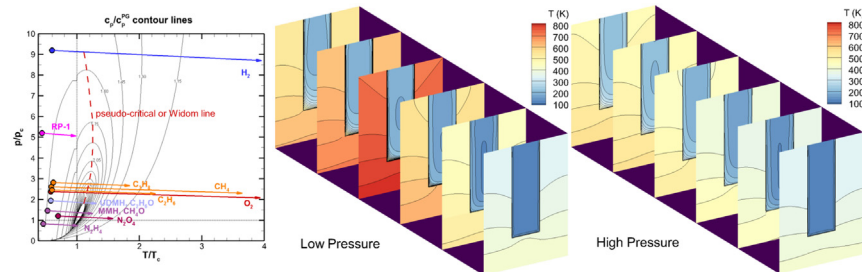
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HIGHLIGHTS

- Supercritical pressure cooling of liquid rocket engines is reviewed and discussed.
- Liquid rocket engine cooling may deteriorate depending on propellant.
- The typical high heat-to-mass flux ratio is the cause for deterioration.
- Deterioration fades out at high supercritical pressures.
- Rough asymmetrically heated channels are less prone to heat transfer deterioration.

GRAPHICAL ABSTRACT



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ABSTRACT

Heating of liquid propellants used as the coolant in rocket engines may lead to undesired phenomena such as pseudo-boiling or heat transfer deterioration under specific conditions. This can be an issue for propellants characterized at the same time by relatively low critical pressure and temperature. Light hydrocarbons, as for instance methane, belong to this family. In the present paper, a critical review is made of the main results obtained by Authors and their coworkers for the present application. Focus is on the correlations and trends inferred by their numerical simulations mainly carried out considering methane as the coolant, perhaps the most challenging one.

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1. Introduction

Liquid propellant rocket engines release a huge amount of power in a relatively small volume. The consequence is that one of the most challenging design issues for this engines is the design

of an effective and efficient cooling system. Aiming to operations in space where no refill from the environment is allowed and no media is available, the rocket designers can only rely in something available on board or on something that must be brought in flight for cooling purposes. Both choices have been explored in the past. For instance, exploiting its unique properties as a coolant, water was used in some development. However, the most successful and effective way of cooling rocket engines has been demonstrated so far that of using a fluid already available on board, possibly without wasting it. In fact, despite the good properties of specific coolants

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Nomenclature

$\mathcal{M}$	molar mass
$\Re$	thermal resistance
c_p	specific heat at constant pressure
D	diameter
D_h	hydraulic diameter
f_w	friction factor
G	mass flux
h	enthalpy
h_c	heat transfer coefficient
k	thermal conductivity
Nu	Nusselt number
p	pressure
Pr	Prandtl number
q_w	heat flux
R	radius
Re	Reynolds number
T	temperature
u	velocity
x	axis

Greek symbols

β	thermal expansion coefficient
δ_v	viscous sublayer thickness
η	distance to wall
μ	viscosity
ρ	density
τ	shear stress
ε	equivalent sand grain roughness

Subscripts

0	referred to $T/T_c = 0.5$ and standard pressure
b	bulk
c	critical
e	exit
i	inlet
o	Dittus-Boelter
t	turbulent
tr	threshold
v	viscous
w	wall

Superscripts

PG	perfect gas
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Table 1

Critical pressure and temperature of common liquid rocket engine propellants.

Propellant (family, name, formula)			p_c (bar)	T_c (K)
Hydrogen	Hydrogen [2]	H ₂	13.0	33.2
Hydrocarbons	Methane [2]	CH ₄	46.0	190.6
	Ethane [2]	C ₂ H ₆	48.7	305.3
	Propane [2]	C ₃ H ₈	42.5	369.8
	Methanol [2]	CH ₃ OH	81.0	512.6
	Ethanol [2]	C ₂ H ₅ OH	61.5	513.9
	RP-1 [3,pp. 182,317]	CH _{1.953}	20.0	678.0
Hydrazines	Hydrazine [2]	N ₂ H ₄	147.0	653.0
	MMH ^a [4]	N ₂ H ₃ (CH ₃)	82.4	567.0
	UDMH ^b [3,p. 317]	N ₂ H ₂ (CH ₂) ₂	60.6	522.0
Oxygen	Oxygen [2]	O ₂	50.4	154.6
Nitrogen Oxides	Nitrogen Tetroxide [2]	N ₂ O ₄	101.0	431.0
	Nitrogen Oxide [2]	NO	64.8	180.0

^a Mono-methyl-hydrazine.^b Unsymmetrical-dimethyl-hydrazine.

of a given thrust level (high specific impulse) and/or to improve engine compactness (high thrust density). Based on this scenario, the liquid rocket engines relying on what has been referred to as “expander cycle” have been developed. Expander cycle engines exploit the power taken by the coolant, typically the fuel, in the cooling system as the input power for turbines. In this way turbines can provide enough power to pumps which in turn provide mechanical energy to fuel and oxidizer which can be therefore fed in the main chamber at high pressure.

This short introduction on the rocket propulsion by liquid propellants is a necessary step for the present discussion, because of the consequence it has on the coolant properties and behavior. We have to distinguish two cases: (1) expander cycles where the coolant is diverted at least in part in turbine beyond the cooling system; and (2) other cycles where the coolant is directed towards the main chamber immediately beyond the cooling system. Two opposite requirement follows. In the first case a thermal machine, i.e. a gas turbine, is required to exploit internal energy, therefore the coolant, entering the cooling system as a liquid, must exit in a gaseous phase, thus experiencing a phase change. In the second case, being cooling more efficient in liquid phase, the requirement is generally to avoid any phase change in the cooling system.

Despite an extremely wide range of possible propellant combination options has been explored [1], only a limited number of propellants have been considered of practical interest. The most important combinations include hydrogen, hydrocarbons and hydrazines as the fuel and oxygen and nitrogen oxides as the oxidizer. If we focus the attention on high performance launcher stages, which are based on pump-fed systems, engine operating conditions typically range between a 40–50 bar for upper stages and up to more than 200 bar for boosters or first stages. According to the common propellant properties reported in Table 1, and considering that coolant pressure in the regenerative cooling system must be greater than pressure in the engine combustion chamber to allow propellant feeding beyond the cooling system, it is evident that in many cases the coolant will be at supercritical pressure, while it may happen that coolant will operate in the cooling system at a pressure close to the critical one.

Once the steady state design condition is reached, hydrogen will always be at supercritical pressure, hydrocarbons and oxygen will be at supercritical pressure for booster/first stage engines, whereas storable propellants like hydrazines and nitrogen oxides will only be at supercritical pressure for very high pressure engines. Let us therefore consider that common values of coolant pressure at the entrance of the cooling system can be assumed to lie in the range

like water, many other fluids can do successfully the job, provided their own properties are taken into account.

On board of most liquid propellant rockets, at least of those used for launcher stages, one can find two fluids, stored in liquid phase: the fuel and the oxidizer that will be burned to generate the high temperature gases necessary to efficient rocket propulsion. The design of the cooling system will therefore rely on the selection and use of one of the propellants (or in principle both in separate cooling systems) as the cooling medium. Such a use has also disclosed the opportunity to recover the combustion-energy collected by the coolant by using it as a pre-heating of the propellant used for cooling. This aspect can be of great importance in the engine system design because it is one of the possible options to avoid the introduction of specific burners or other complex devices to get the hot gases. Hot gases are required to feed turbines in turn driving pumps needed to feed propellants at high pressure in the main chamber. Getting high pressure is one of the goals of the rocket designer as it allows to reduce propellant consumption for the generation

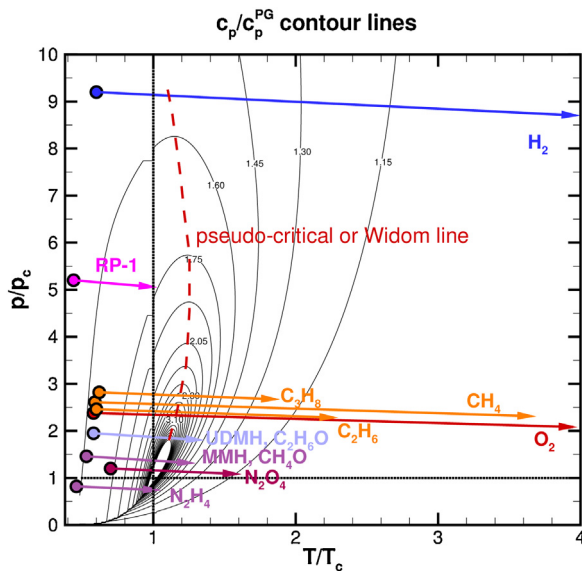


Fig. 1. Qualitative propellant evolution in cooling channels assuming an inlet pressure of 120 bar and an exit temperature of 700 K.

between 50 and 400 bar for pump-fed engines. To highlight the different conditions of possible coolants let us roughly compare their evolution in a state diagram, for a specific condition, say with inlet pressure of 120 bar. For this purpose it is particularly convenient to resort to a reduced-temperature (T/T_c) reduced-pressure (p/p_c) state diagram. As known the reduced quantities are not exactly the same for the different possible coolants. Yet, plotting the isocontours of the ratio between specific heat at constant pressure of the real fluid and the value of perfect gas, for instance for methane, gives a qualitative behavior of the displacement from perfect gas for the different coolants (see Fig. 1). A maximum allowable coolant temperature of 700 K is considered in the diagram without taking into account possible lower maximum allowable temperature due to propellant stability or other intrinsic risks. In these conditions, it is evident that two extreme cases can be found. From the one hand if coolant is hydrogen its reduced pressure is so high (about 10) that its properties will smoothly evolve from liquid-like to gas-like and then perfect gas conditions. Note that the hydrogen arrow to reach 700 K would continue up to a reduced temperature of 21. On the other hand, one should consider the cases of hydrazine and RP-1. Despite the quite different critical pressure, they are hardly able to exceed their critical temperature (see Table 1), therefore they will certainly behave as liquid coolants. It is instructive to see (Fig. 2) that if a modest temperature increase of the coolant is considered, most of propellants lie in the liquid or liquid-like phase, with the exception of the following three cases: hydrogen, methane and oxygen. Hydrogen is still a singular case as it gasifies also with a little temperature increase. By all means, it will cross the pseudo-critical or the saturation temperature: pseudo-phase change will always occur if pressure is more than 13 bar. Methane and oxygen behave similar to each other due to the similar values of critical temperature. It is important to underline that, for these coolants, also a modest heating bring them to a temperature close to the critical one with the possible occurrence of a phase change or a pseudo-phase change if pressure is greater than 50 bar.

Based on the above discussion, it is not surprising that studies on the possible occurrence of pseudo-phase change in rocket cooling channels have flourished recently, when the attention to the use of methane as a rocket fuel has more and more increased. The interest towards methane comes from its possible use as a denser and cheaper replacement of hydrogen in launch vehicles [5,6] and

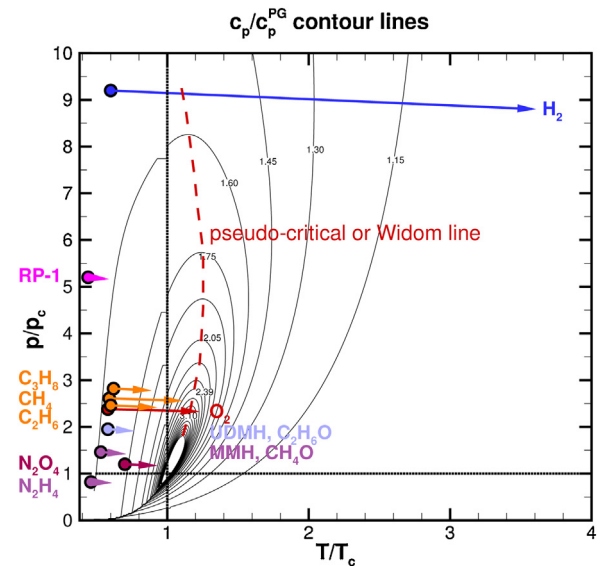


Fig. 2. Qualitative propellant evolution in cooling channels assuming an inlet pressure of 120 bar and a temperature increase of 100 K.

as a cheaper replacement of toxic storable propellants for space propulsion [7,8].

Methane has been proposed for different kind of engines including those using it as the coolant in the regenerative cooling systems with following injection in the combustion chamber and those based on expander cycles which exploit gaseous methane as the working fluid of a gas turbine. It has to be mentioned that there are also studies which foresee the use of oxygen as the coolant and possibly as the turbine gas. In the past, oxidizers have only been considered as coolant for little temperature increase due to the less efficiency as coolants than fuels and to concerns about the occurrence of oxidation phenomena under heating. Nevertheless, because of the improved knowledge of material design and behavior the use of oxygen cooling for an expander cycle is currently being studied [9].

In the present paper, we present a review of the main phenomena involved in the propellant heating in cooling channels with focus on “near-critical” conditions. The discussion relies on the many numerical simulations made by the Authors that without lack of generality mainly addressed the case of methane heating. The paper is organized as follows. In Section 2 the general phenomena related to phase change in sub and supercritical pressure conditions is presented with focus on the possible interest and caution related to its occurrence in rocket cooling channels. Section 3 discusses a general approach to estimate a priori the possible occurrence of detrimental heat transfer deterioration in uniformly heated round tubes. Section 4 discusses a general approach to estimate a priori the temperature evolution under heat transfer deterioration in uniformly heated round tubes. Section 5 presents the consequences of asymmetric heating on the development of heat transfer deterioration phenomena. In Section 6 the mitigation of the heat transfer deterioration phenomenon caused by wall roughness is discussed. Eventually, a summary of the main conclusions achieved is reported.

2. Propellant heating and (pseudo-)phase change in rocket cooling channels

At subcritical pressures, coolant boiling is – per se – an extremely effective cooling mean, as the coolant is able to cool the wall while keeping its bulk temperature unchanged as the incoming heat is accumulated as the heat of vaporization required for the phase

change. For this reason, the so called nucleate boiling is considered as a possible option in the most thermally solicited point of a liquid rocket engine cooling system, that is typically close to the nozzle throat [3]. However, to exploit nucleate boiling, suitable conditions for the coolant temperature and pressure, as well as for local and overall heating, have to be satisfied. In general, boiling in channels can occur in different ways depending on flow and heating conditions [10]. It is important to underline here some aspects relevant to subcritical boiling that can be of help in understanding phenomena at supercritical pressure. If one considers a liquid flowing in a heated straight channel, wall temperature may exceed the saturation temperature at a given position, originating its boiling, while the liquid remains subcooled, that is at a temperature lower than the saturation one. If the heat flux is reasonably low, except than in a limited region along the channel, it may happen that nucleate boiling occurs in that region, with a significant enhancement of convective heat transfer, while the liquid bulk temperature remains below the saturation temperature along the whole channel. On the other hand, if the flow is heated at a constant rate (i.e. uniform heat flux along the tube), bulk temperature will continue to increase up to reach and exceed in turn the saturation temperature. It is here that different scenarios appear. A detailed description is reported in [10] and is out of the scope of the present paper. It is worth, however, to remind that at high heat fluxes (as those often characterizing liquid rocket engine cooling channels), by continuous heating and while liquid phase is still subcooled, a condition called “departure from nucleate boiling” occurs, similar to the critical heat flux condition in pool boiling [11], with the generation of a gaseous film along the tube walls. The occurrence of the “film boiling” appears when a critical heat flux level is exceeded and is a dangerous phenomena because an abrupt deterioration in the cooling process takes place. The reason for this phenomenon in a forced convective flow is essentially similar to that observed in pool boiling: it originates by the insulating vapor film which covers the heated walls which is characterized by the quite lower thermal conductivity of the vapor as compared to that of the liquid and especially impairs the turbulent exchange between the wall and the core flow. It is also worth noticing that despite the different properties (viscosity, specific heat and thermal conductivity) between liquid and gas phase, coolant capabilities are not dramatically different for the gaseous or liquid single-phase flow if one considers the same channel and mass flow rate. This is due to the fact that mass flux does not change and the gas is less dense but flows at higher velocity than the liquid. The remaining differences, which are due to the different properties (viscosity, specific heat and thermal conductivity), are in great part compensated. Therefore, more than the poorer performance of the gas as the coolant as compared to the liquid, what discourages choosing to have a phase change in a LRE cooling channels is the phase-change itself where a local, abrupt heat transfer deterioration due to film boiling may occur.

2.1. Qualitative analysis: comparison with subcritical phase change

The short review of the importance of phenomena as heat transfer enhancement (i.e. nucleate boiling) and deterioration (i.e. film boiling) occurring in pool boiling experiments because of subcritical phase change, can be of help for a first qualitative analysis of the coolant flow behavior when operating at supercritical pressure. A more detailed analysis will be carried out in the next Section 2.2, where the effects of forced convection will be discussed. Keeping in mind the possible trajectories traced in the reduced-pressure/reduced-temperature diagram of Fig. 1, different modes of heat transfer can occur also in the supercritical regime similarly to what happens in the subcritical regime. According to the different behavior occurring if the heat flux is less or more than

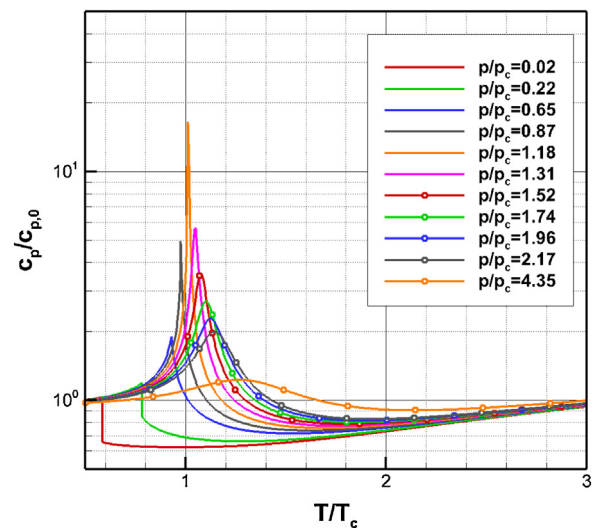


Fig. 3. Methane specific heat variation with temperature at different pressures. c_p is nondimensionalized with its value, $c_{p,0}$, at $T/T_c = 0.5$ and standard pressure. Data from [15].

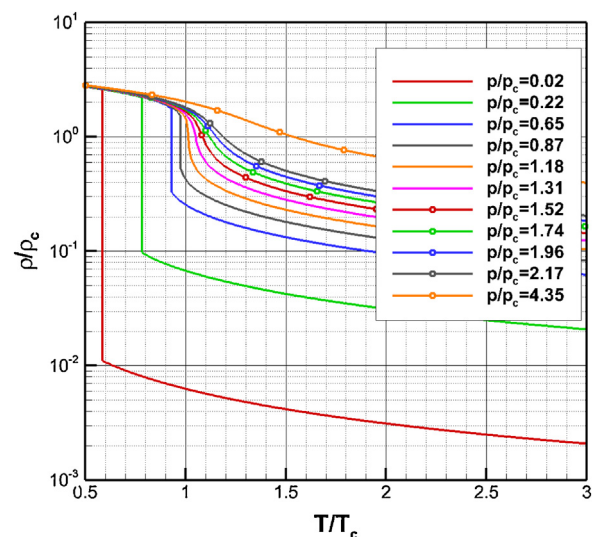


Fig. 4. Methane density variation with temperature at different pressures is nondimensionalized with its value at $T/T_c = 0.5$ and standard pressure. Data from [15].

the critical heat flux at subcritical pressure, if one would extrapolate a similar behavior at supercritical pressure, he should find heat transfer enhancement or deterioration depending on the heat flux level. While the physics underlying subcritical and supercritical processes is quite different, because the former is characterized by discontinuous changes while the latter is characterized by a spread and continuous transition [12], their comparison has been widely followed in the literature (see for instance [13,14]) with different conclusions about similarities and differences. What is important to underline here is that experience leads to the conclusion that there is a heat transfer enhancement when the heated fluid approaches the Widom line and that a further heat flux and wall temperature increase can lead to heat transfer deterioration.

For the present qualitative analysis we can justify the occurrence of heat transfer enhancement, taking advantage from plots of specific heat, density and thermal conductivity as a function of pressure and temperature for methane (Figs. 3–5). Actually, at subcritical pressure the single methane flow particle slightly improves heat exchange because it increases its specific heat (Fig. 3). Then, once it

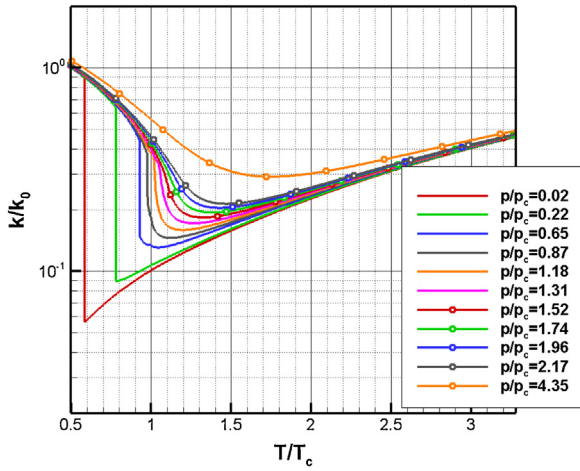


Fig. 5. Methane thermal conductivity variation with temperature at different pressures. Thermal conductivity is nondimensionalized with its value, k_0 , at $T/T_c = 0.5$ and standard pressure. Data from [15].

reaches its saturation temperature, it vaporizes yielding a far better heat exchange promoted by the heat transferred to the bulk of the coolant by bubble diffusion and re-condensation. At supercritical pressure, there will no longer be the vaporization phenomenon and this removes the main reason for heat transfer enhancement. Nevertheless, at least at near-critical conditions, i.e. when pressure is not much larger than critical pressure, the increase of specific heat is quite larger than at subcritical conditions (Fig. 3). The high increase of specific heat may produce a significant heat transfer enhancement at near critical conditions. According to data reported in Fig. 3, it is also evident that the specific heat increase softens as higher pressures are considered. Therefore, it is not surprising that heat transfer enhancement is not detected at pressure much higher than critical. The decrease of specific heat occurring, under heating, as temperature crosses the Widom line would lead to a reduction of heat transfer that can be considered as a return to normal heat transfer after the enhancement phase. Note that heat transfer enhancement is also referred to as improved heat transfer [13].

The heat transfer deterioration at subcritical pressures occurs at the critical heat flux with the transition from nucleate to film boiling due to the saturation of the boiling process and to the corresponding formation of an insulating gaseous layer on the heating wall. At low supercritical pressure, the near critical regime, similarities with the subcritical regime are remarkable [14]. Also, at low supercritical pressure the high heat transfer rate can generate a low-conductivity/low-density layer which impedes the exchange of heat with fresh colder fluid particles. In a pool-boiling analogy, this can be referred to as pseudo-film-boiling and its occurrence may lead to a significant deterioration of heat transfer. It can be easily inferred that this phenomenon would be directly related to an abrupt change of flow density with temperature, indicating the presence of a pseudo-gas layer featuring low thermal conductivity. Looking at Fig. 4, and considering that both density and thermal conductivity variations through the Widom line soften at high reduced pressures, Fig. 5, it can be promptly seen that such a heat transfer deterioration will more likely occur in near-critical conditions rather than far from the critical pressure.

2.2. Fluid dynamic analysis

As discussed above, in the near-critical region, the effectiveness of heat transfer to a fluid is significantly influenced by the amount of entering heat flux. Differently from pool boiling, in forced con-

vection, heat transfer is influenced by the ratio of entering heat flux to mass flux, q_w/G , similarly to the subcritical region. In particular, at relatively high mass flux or low heat flux the heat transfer may be enhanced, while at relatively high heat flux or low mass flux a heat transfer deterioration may occur.

In this section a discussion of why these quantities play a role in heated supercritical flow in channels is presented. The heat transfer deterioration phenomenon is originated, as in the pool boiling, by the formation of a low-density low-conduction layer at wall. However, when considering a turbulent flow in a tube, turbulent diffusion plays a decisive role in determining the possible occurrence of heat transfer deterioration. Turbulent diffusion has therefore to be studied in detail. In pursuing this study, it is useful to consider that, as the typical flow conditions in rocket engine cooling channels are relevant to relatively high mass flux, of the order of thousands of kg/s/m^2 , the resulting turbulent flow can be considered unaffected by any buoyancy effect.

To investigate the flow structure under heat transfer deterioration, the turbulent flow inside a heated tube is considered to be composed of two regions [16,17]: the tiny viscous sublayer close to the wall where the turbulent shear stress τ_t is negligible and the above turbulent core of the flow where the viscous shear stress τ_v is negligible. According to this modeling, the heat transfer coefficient of the flow can be expressed as $h_c = 1/(\mathfrak{R}_v + \mathfrak{R}_t)$ where $\mathfrak{R}_v$ and $\mathfrak{R}_t$ are the thermal resistances of the two regions. For the heat transfer deterioration to occur a sufficiently large increase of $\mathfrak{R}_v$ and/or $\mathfrak{R}_t$ should take place. The thermal resistance of the viscous sublayer can be estimated as:

$$\mathfrak{R}_v = \frac{\delta_v}{k_w} \quad (1)$$

where δ_v is the viscous sublayer thickness and k_w is the fluid thermal conductivity at wall; note that, because of the very tiny thickness of the viscous sublayer, the value of k_w is representative of the thermal conductivity across the whole viscous sublayer. The thermal resistance of the turbulent core is estimated as $\mathfrak{R}_t = R/k_{t,b}$ where R is the tube radius and $k_{t,b}$ is the bulk turbulent thermal conductivity which can be related to the turbulent viscosity μ_t by means of the turbulent Prandtl number $Pr_t = \mu_t c_p / k_t$. Note that, accordingly to the hypothesis of sufficiently large Reynolds number, almost the entire section of the tube is dominated by turbulence and thus the tube radius R is the thickness of the turbulent core. The turbulent thermal resistance is then:

$$\mathfrak{R}_t = \frac{R Pr_t}{\mu_{t,b} c_{p,b}} \quad (2)$$

where the turbulent Prandtl number is generally considered constant. According to Eq. (2), a reason for heat transfer deterioration is found in a possible reduction of μ_t in the turbulent core of the flow-field. The behavior of the turbulent viscosity can be easily associated to that of the shear stress τ , which is defined as:

$$\tau = \tau_v + \tau_t = (\mu + \mu_t) \frac{\partial u}{\partial \eta} \quad (3)$$

where μ is the viscosity, u is the velocity component along the channel streamwise axis, and η is the distance to wall. Finally, the relation between the turbulent viscosity and the shear stress results to be $\mu_t \sim \sqrt{\tau}$ since in the core region $\tau_v \ll \tau_t$ and, according to the Prandtl's mixing-length theory, the turbulent viscosity is related to the partial derivative of the streamwise velocity $\partial u / \partial \eta$. Following the same approach used in [18], the shear stress close to the wall can be expressed as:

$$\tau = \tau_w \left(1 - \frac{\eta}{R} \right) - G \frac{du_b}{dx} \eta \quad (4)$$

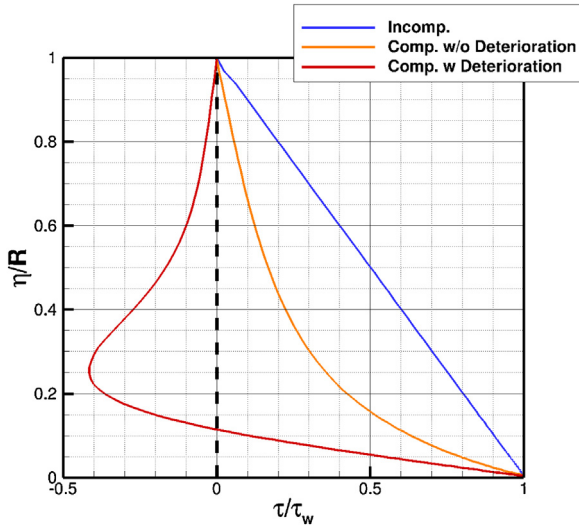


Fig. 6. Schematic of shear stress profile for different fluids and flow regimes.

where τ_w is the wall shear stress, G is the mass flux, u_b is the bulk velocity, and x is the streamwise direction. Eq. (4) provides a link between the shear stress in the vicinity of the wall and bulk properties. It shows that, for a given mass flux G , in case of fluid acceleration (i.e. positive du_b/dx) the shear stress τ falls more rapidly with distance from the wall than it would in the case of constant bulk velocity. Eq. (4) highlights that shear stress reduction in the vicinity of the wall cannot occur as long as the flow is in the liquid phase (i.e. it can be considered incompressible) because in that case u_b is constant (i.e. it comes from the mass conservation principle for a constant cross section tube), whatever is the incoming heat transfer rate.

The typical shear stress behavior in a circular cross-section channel with the numerical model developed in [19] is shown in Fig. 6 in non-dimensional form for different fluids and flow regimes. In case of an incompressible fluid with constant properties, the shear stress has a linear trend, from $\tau = \tau_w$ at wall to $\tau = 0$ at the tube axis. Different is the case of a compressible fluid. In fact, bulk velocity changes according to the bulk density reduction, which is the result of the pseudo-boiling. The higher the heat transfer rate, the higher will be the pseudo-gas production rate at the wall, and as a consequence the bulk density reduction and bulk velocity increase. Therefore, according to Eq. (4), a high heat transfer rate, once the pseudo-boiling phenomenon is taking place, implies a shear stress reduction, up to negative values, and thus a major loss of turbulent energy. It is interesting to compare the results obtained at different heat transfer rates. If one considers moderate heat transfer a shear stress reduction occurs because $du_b/dx > 0$, but it is not sufficient to cause negative values. The behavior of the shear stress has an evident influence on the velocity profile as shown in Eq. (3); for instance, when the shear stress is zero the velocity presents a peak value. The velocity profiles corresponding to the shear stress distributions shown in Fig. 6 are presented in Fig. 7 in non-dimensionalized form with respect to the velocity at the symmetry axis ($\eta = R$). According to this point of view, when the heat transfer rate to a supercritical fluid is large enough, the flow velocity increases near the wall more than in the high-density core region: the velocity profile is modified in such a way that large region of low shear stress are present and thus heat transfer deterioration occurs.

Once the occurrence of velocity peak has been discussed on the basis of qualitative arguments and numerical solutions, the subsequent step is to identify under which heat flux level this

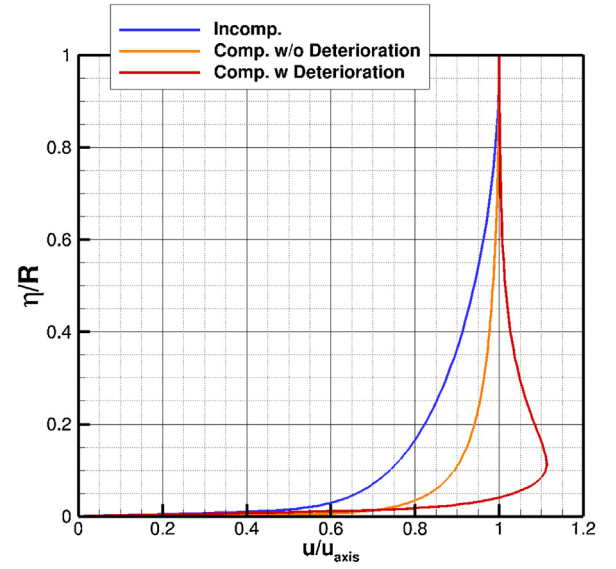


Fig. 7. Schematic of velocity profile for different fluids and flow regimes.

phenomenon arises. To this goal Eq. (4) can be further developed including the integral fluid energy equation:

$$\tau = \tau_w \left(1 - \frac{\eta}{R}\right) - 2q_w G \left(\frac{\beta}{\rho c_p}\right)_b \frac{\eta}{R} \quad (5)$$

where q_w is the entering heat flux,

$$\beta = -\frac{1}{\rho} \left(\frac{\partial \rho}{\partial T}\right)_p \quad (6)$$

is the thermal expansion coefficient, ρ is the density, and the subscript b in the parenthesis means that all the variables inside are evaluated at the bulk state. Details of the derivation of the above Eq. (5) are given in [18]. The abscissa at which the shear stress reaches the zero value is:

$$\left(\frac{\eta}{R}\right)_{\tau=0} = \left[1 + \frac{2q_w G}{\tau_w} \left(\frac{\beta}{\rho c_p}\right)_b\right]^{-1} \quad (7)$$

This shows that the larger the non-dimensional term

$$\frac{q_w G}{\tau_w} \left(\frac{\beta}{\rho c_p}\right)_b \quad (8)$$

the smaller the abscissa $(\eta/R)_{\tau=0}$ and thus the stronger the turbulent energy depletion. In other words, there is a threshold value for the above non-dimensional term that, if exceeded, implies heat transfer deterioration. Introducing the definition of the friction factor, $f_w = 4\tau_w / \frac{1}{2}\rho_b u_b^2$, it is possible to identify the onset of heat transfer deterioration as:

$$\frac{q_w}{G f_w} \left(\frac{\beta}{c_p}\right)_b > K \quad (9)$$

where K is the threshold value. This result is essentially similar to that proposed in [20] and equal to that proposed in [21] where the threshold value is also estimated to be $K = 0.187$. Eq. (9) shows that heat transfer deterioration occurs for large heat flux q_w and low mass flux G . Moreover, for a given flow in a tube, heat transfer deterioration occurs in the tube sections where the term $(\beta/c_p)_b$ is sufficiently large. The behavior of the thermodynamic variable β/c_p for a generic supercritical fluid is presented in Fig. 8 in non-dimensional form. This figure shows that, for a given supercritical pressure, β/c_p presents a maximum which nearly occurs at the pseudo-critical temperature. Increasing the pressure level, the

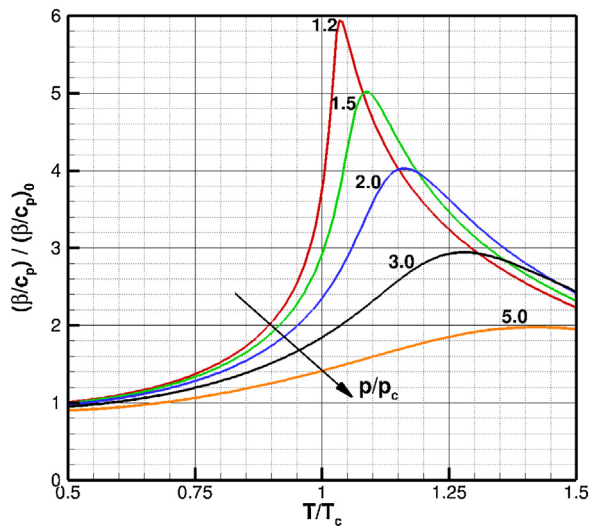


Fig. 8. Thermodynamic variation of β/c_p with temperature and for different values of supercritical pressure, in non-dimensional form. β/c_p is nondimensionalized with its value, $(\beta/c_p)_0$, at $T/T_c=0.5$ and standard pressure.

maximum decreases and occurs at larger temperature. The maximum almost disappears when the pressure level is sufficiently high; in fact, for $p/p_c > 5$ a maximum is hardly identified. Moreover, looking at the behavior of β/c_p in Fig. 8, it is also possible to recognize that when deterioration takes place a further heating has a positive effect because the fluid temperature exceeds the pseudo-critical value and thus the variable β/c_p decreases. As a consequence, the heat transfer capability of the flow is restored because of the increase of the shear stress, Eq. (5). Although the above theoretical analysis is based on many simplifying assumptions that may reduce the quantitative goodness of the model, the general finding is rather solid. It is now evident that, for a supercritical fluid entering in a tube with a subcritical temperature, the deterioration occurs when the fluid bulk temperature is in the pseudo-critical region, provided that the mass flow rate is sufficiently small and the entering heat flux is sufficiently large. Finally, deterioration is stronger as the fluid pressure level is close to the critical value because of the relatively large value of β/c_p .

The above theoretical analysis is based on the assumption that heat transfer deterioration occurs when the turbulent thermal resistance $\mathfrak{R}_t$ sufficiently increases. However it is only a part of the reason for heat transfer deterioration because the wall temperature increase, caused by the increase of $\mathfrak{R}_t$, implies also an increase of the thermal resistance of the viscous sublayer, $\mathfrak{R}_v$. The wall temperature increase, in fact, implies a thickening of the viscous sublayer which more than offsets the increase of the thermal conductivity k_w with T_w , resulting in an increase of $\mathfrak{R}_v$ according to Eq. (1). The increase of the thermal conductivity with temperature is due to the gaseous behavior of the fluid close to the wall because of the relatively high temperature of the wall when deterioration takes place. Here, it is useful to note that because of turbulent diffusion the role of thermal conductivity is confined to the viscous sublayer where temperature can be high enough to be far from the low values expected at the Widom line crossing.

The two thermal resistances, $\mathfrak{R}_t$ and $\mathfrak{R}_v$, can be of the same order of magnitude because $k_t \gg k_w$ while $\delta_v \ll R$. Fig. 9 presents the qualitative behavior of the two thermal resistances for the deteriorated case already shown in Figs. 6 and 7. Also the behavior of the wall temperature, non-dimensionalized with the critical temperature, is presented in that figure to better appreciate the region where the heat transfer deterioration takes place. In Fig. 9 the thickness of the viscous sublayer δ_v has been evaluated as the wall

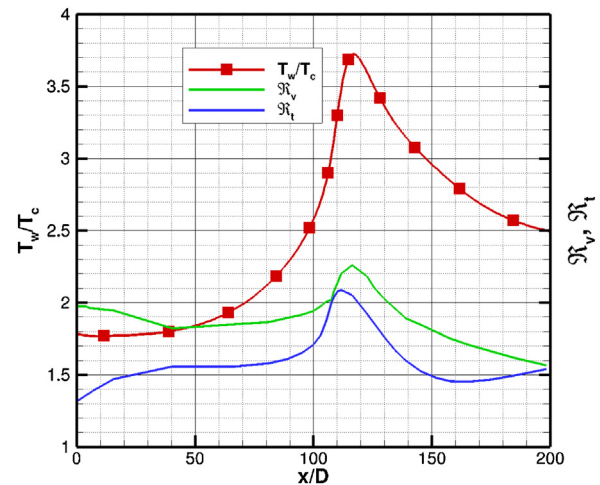


Fig. 9. Schematic of wall temperature and thermal resistances of the viscous sublayer and turbulent core, in non-dimensional form.

distance where the turbulent shear stress is half the value of the total shear stress: $\tau_t/\tau = 0.5$. Note that the fraction τ_t/τ adopted to define δ_v does not influence much the qualitative finding of Fig. 9, that is, the thermal resistances $\mathfrak{R}_v$ and $\mathfrak{R}_t$ can be of the same order of magnitude and both increase with β/c_p . Ultimately, the heat transfer deterioration in case of near critical fluids is related to both the decrease of the turbulent energy in the core region of the flow and the thickening of the viscous sublayer close to wall.

The present discussion shows that, differently than in the case of pool pseudo-boiling, the occurrence of heat transfer deterioration in turbulent forced convection is linked to the properties of turbulent diffusion and heat exchange.

3. Prediction of heat transfer deterioration phenomena in straight tubes

One important aspect in designing cooling channels to be fed by a near-critical coolant is the preliminary evaluation of the possible occurrence of heat transfer deterioration. Of course, similarly to film-boiling, also heat transfer deterioration is to be avoided, because the corresponding wall temperature increase may lead to wall material failure. To better focus on the possible problem let us consider which are the necessary conditions for the phenomenon to occur. For heat transfer deterioration to occur the pseudo-gas phase must develop on the surface. This is possible if wall temperature is greater than the pseudo-critical temperature. Moreover, obviously the coolant bulk temperature has to be lower than the pseudo-critical temperature else we would be in the gas-like regime. The condition that the pseudo-critical temperature is between the wall and bulk temperature is a necessary but not sufficient condition because, as underlined before, the coolant pressure as well as the heating rate will determine if heat transfer deterioration takes place.

To determine if the heat transfer process can be considered deteriorated or not it is necessary to identify a deterministic criterion for its occurrence. Different criteria can be found in the literature and exhaustive reviews of them have been recently published [22,23]. In the literature, the criterion often refers to the normal heat transfer conditions and identifies a deteriorated heat transfer as that occurring at a lower rate than in normal conditions. The uncertainty of this definition is that it assumes as known the normal heat transfer coefficient value in the conditions where heat transfer deterioration is to occur. Often the normal heat transfer evaluation is arbitrarily based on the Dittus-Boelter relation. On the other hand, it is important to identify the possible occurrence

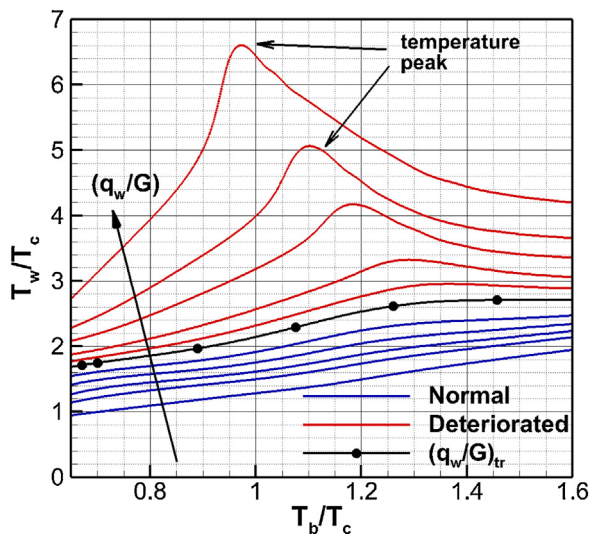


Fig. 10. Wall temperature evolution in a channel cooled by methane at a reduced pressure of 2.8 and different values of heat to mass flux ratio. (For interpretation of the references to color in the text, the reader is referred to the web version of this article.)

of heat transfer deterioration from the mentioned reasons, namely the wall temperature increase up to a maximum value, caused by the poor heat exchange with the coolant. Therefore, in [24] a clearer deterministic criterion is introduced. According to that criterion, in the following the heat transfer is said to be deteriorated if a temperature peak takes place along the channel.

Once the criterion is set, it is useful to rely on the known main dependencies recalled in Section 2.2 to identify the conditions for the occurrence of heat transfer deterioration. Key parameters have been found as the heat flux to specific mass flow rate ratio and the reduced pressure. Aiming to mimic possible conditions of cryogenic propellants at cooling channel entrance, a parametric analysis has been carried out in [24] assuming that the entrance temperature in the tube is 5% higher than the standard evaporation temperature and with an inlet Reynolds number of about $3 \cdot 10^5$. The ratio q_w/G at a given reduced pressure is parametrically changed to identify the threshold value for the occurrence of heat transfer deterioration. The detailed discussion is reported in [24], however it is worth highlighting some results that will be useful in the following.

In particular, Fig. 10 shows the wall temperature evolution and Fig. 11 the convective heat transfer coefficient evolution along the channel at reduced pressure of 2.8 for methane. Note that the abscissa is replaced by the more significant bulk temperature, which increases with abscissa because of the continuous heating along the channel.

The threshold value $(q_w/G)_{tr}$ is unambiguously identified with the present criterion. In fact, by the analysis of wall temperature derivative with respect to abscissa (or bulk temperature) it can be easily seen if its value changes sign or not, and so the occurrence or not of a tiny or large temperature peak that, on the basis of the present criterion, identifies heat transfer deterioration. Fig. 10 shows the temperature behavior at the threshold value as identified by a black line. It can be clearly seen that for heat to mass flux ratio greater than the threshold value $(q_w/G)_{tr}$, see the curves in red, a temperature peak occurs at a given bulk temperature, whose value lowers as (q_w/G) increases. On the other hand, the wall temperature curves at lower than threshold heat-to-mass-flux ratio, highlighted in blue, show a normal behavior with a reasonably smooth increase with bulk temperature. The behavior of heat transfer coefficient is less clear (Fig. 11) in the lower than threshold heat-to-mass-flux ratio with an increasing/decreasing behavior up to tend towards a

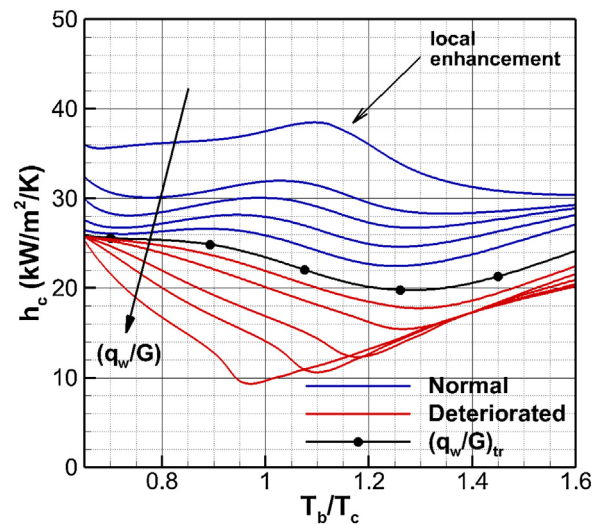


Fig. 11. Heat transfer coefficient evolution in a channel cooled by methane at a reduced pressure of 2.8 and different values of heat to mass flux ratio. (For interpretation of the references to color in the text, the reader is referred to the web version of this article.)

final value when flow has reasonably reached a fully gaseous phase. On the other hand, the deteriorated cases are exactly those showing a clear heat transfer coefficient decrease up to a minimum. Despite it seems that a criterion for the evidence of heat transfer deterioration could have been set also on the basis of the heat transfer coefficient plot, the presence of possible local enhancement (see the local h_c peak in the curves in blue) makes identification not straightforward. In fact, criteria based on heat transfer coefficient are based, as mentioned above, on a comparison with normal heat transfer coefficient value.

It has to be said that, despite the criticism on criteria relying on the validity of Dittus-Boelter relation, present results are in line with different criteria available in the literature. The latter, however, are not always able to identify as deteriorated some conditions with heat transfer deterioration shown in Fig. 10. A commonly considered criterion [25] based on the ratio between heat transfer coefficient and the value obtained by Dittus-Boelter relations $h_c/h_{c,0} < 0.3$ would exclude some of the deteriorated cases as shown in Fig. 12. Note that the a similar criterion proposed in [26] with $h_c/h_{c,0} < 0.5$ would correctly identify the threshold as considered in the present criterion.

Based on this criterion a correlation was found in [24] for coolants entering a straight tube with constant heat flux at a temperature which is 60% the critical temperature and considering different coolants. A general dependence was found as follows:

$$(q_w/G)_{tr} = 275 + 4.975 (\mathcal{M} + 23.6) [p_i/p_c - 1.84 + 3.4(c_p/\beta)_c] \quad (10)$$

where the threshold value in J/kg is obtained considering the molar mass $\mathcal{M}$ in kg/kmol and the ratio $(c_p/\beta)_c$ in MJ/kg. This correlation has been developed for a typical rocket cooling channel setup, and considering three light hydrocarbons and a wide range of inlet reduced pressure p_i/p_c . However, its validity could be extended with further tests on different coolants. It provides a linear dependency of the threshold heat flux for heat transfer deterioration on reduced pressure for each coolant and it is useful to highlight here that the dependence on the specific coolant is found through two parameters: the molar mass $\mathcal{M}$ and the ratio $(c_p/\beta)_c$. The importance of this latter parameter was already discussed above. Here it is worth to underline that the value of the ratio c_p/β at critical conditions is a finite value (see Table 2), which only depends on the considered substance.

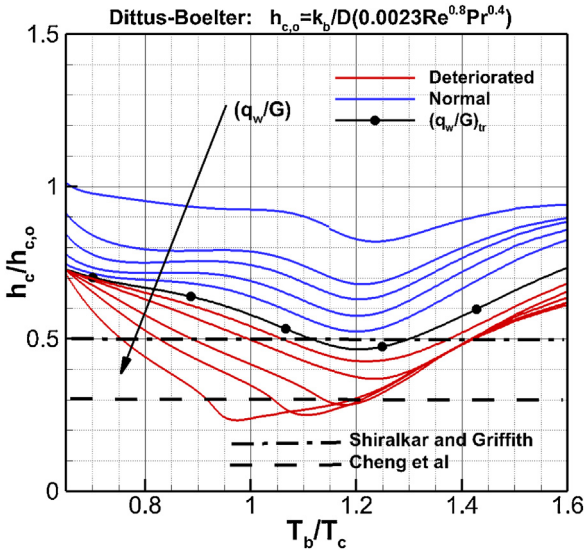


Fig. 12. Heat transfer coefficient and comparison of criteria for the occurrence of heat transfer deterioration.

Table 2

Coolant properties.

Coolant	p_c [MPa]	T_c [K]	$\mathcal{M}$ [kg/kmol]	$(c_p/\beta)_c$ [MJ/kg]
Methane	4.5992	190.564	16.042	0.1702
Ethane	4.8718	305.322	30.069	0.1530
Propane	4.2477	369.825	44.096	0.1323

4. Evaluation of heat transfer when deterioration phenomena occur

Once the conditions that trigger heat transfer deterioration in supercritical fluids have been identified, it is useful to derive appropriate correlations for the Nusselt number in order to evaluate crucial parameters like wall temperature and heat flux quickly, at least as far as it can be accepted to consider devices operating at mildly deteriorated conditions. Correlations for quick evaluation are particularly useful during design phase, when massive parametric analysis are required. Nusselt number correlations have been usually determined on the basis of experimental data. However, with today capabilities, they can also be calibrated on the basis of numerical simulations. As an example we report here results from such an approach which has been used in [18] to derive a Nusselt number correlation for supercritical methane in straight tubes with round section under heat transfer deterioration and with negligible buoyancy effects. In that approach the numerical solutions of the Reynolds averaged Navier–Stokes equations have been efficiently used to investigate trends of the heat transfer to supercritical fluids under a sufficiently wide range of operative conditions, circumventing the intrinsic difficulty of the experimental apparatus in exploring the same wide range of operative conditions (in terms of mass flux, heat flux, fluid pressure and temperature, etc.). The result achieved in [18] is summarized by the correlation:

$$Nu_b = 0.026 \cdot Re_b^{0.8} Pr_b^{0.16} \Lambda^{0.28} \quad (11)$$

which corrects the classical Dittus–Boelter relation for the exponent of the bulk Prandtl number and for the additional term Λ which is a function, for a given fluid pressure, of the wall and the bulk temperature. More specifically Λ is defined as:

$$\Lambda = \frac{\rho_w}{\rho_b} \frac{k_w}{k_b} \frac{\bar{c}_p}{c_{p,b}} \left(\frac{\mu_b}{\mu_w} \frac{T_b}{T_w} \right)^2 \quad (12)$$

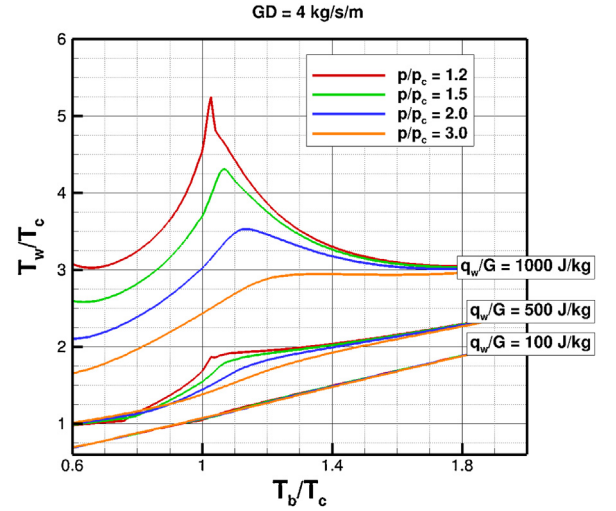


Fig. 13. Reduced wall temperature behavior of a methane flow having $GD = 4$ kg/s/m.

where $\bar{c}_p$ is the averaged constant pressure specific heat over the cross-section of the tube, which is defined as $\bar{c}_p = (h_w - h_b)/(T_w - T_b)$. The correlation was obtained on the basis of a regression of numerical solutions of deteriorated flows in tubes with a diameter of $D = 1$ mm and the following conditions. The inlet temperature was in the range $T_i/T_c = 0.6 - 1.0$. The exit pressure was in the range $p_e/p_c = 1.2 - 3.6$. The mass flux was in the range $G = 4000 - 8000$ kg/s/m². The resulting inlet Reynolds number was therefore in the range $3 \cdot 10^4 - 3 \cdot 10^5$.

Once the Nusselt number is evaluated through Eq. (11), the wall temperature can be easily obtained as:

$$T_w = T_b + \frac{q_w/G}{c_{p,b}} \cdot \frac{Re_b Pr_b}{Nu_b} \quad (13)$$

The above equation highlights that the wall temperature is a function of the fluid bulk temperature and pressure, the ratio of the heat-flux to mass-flux, and the product of the mass-flux for the tube diameter, GD . The latter dependency can be recognized through the definition of the Reynolds number, $Re_b = GD/\mu_b$. Here it is worth noting, as already done in [18], that the correlation of Eq. (11) can mispredict the peak wall temperature with an error up to 200K when the fluid pressure is rather close to the critical point (i.e. $p_e/p_c = 1.2$) even if the Nusselt number error is below 20%. However, for larger pressure levels the peak wall temperature is predicted much more accurately. Even considering the issue of correctly estimating the wall temperature peak when the fluid pressure is close to the critical value, Eq. (11) can be successfully used to evaluate the parameters that impact most the heat transfer deterioration. Of course, such analysis is the more reliable the more the parameters are varied in the same range of those used for the calibration of Eq. (11). The behavior of the wall temperature as a function of the bulk temperature (both non-dimensionalized with the critical temperature) of a methane flow having $GD = 4$ kg/s/m (that is, with an inlet Reynolds number of about $3 \cdot 10^4$ when $T_i/T_c = 0.6$) and at different values of the reduced pressure p/p_c and the heat-to-mass flux q_w/G is presented in Fig. 13. As discussed in the previous sections, it is easy to recognize that heat transfer deterioration is possible only after a threshold value of the heat-to-mass flux and that deterioration is more severe the more the fluid pressure is closer to the critical pressure.

Fig. 13 shows that, for $q_w/G = 100$ J/kg the wall temperature is only slightly higher than the bulk temperature and there is no evidence of the pressure effect, whereas deterioration is achieved when $q_w/G = 1000$ J/kg for all values of reduced pressure, although

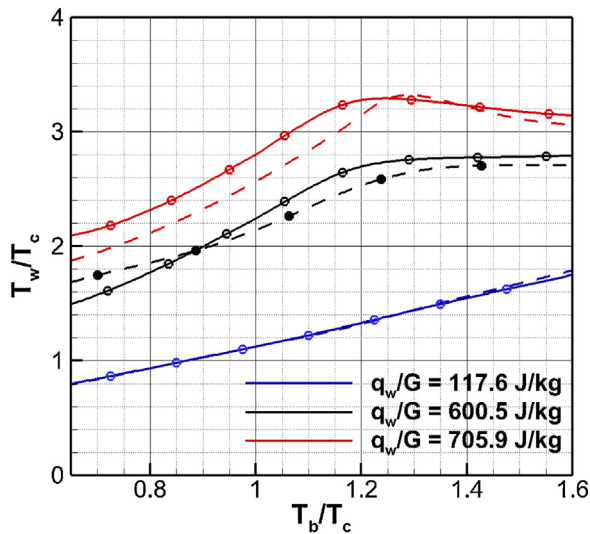


Fig. 14. Reduced wall temperature behavior of a methane flow with $GD = 34 \text{ kg/s/m}$ and $p/p_c = 2.8$. Comparison of results of Nusselt correlation [18] (solid lines) and numerical solutions [24] (dashed lines).

almost indiscernible at $p/p_c = 3$, as expected according to threshold condition given in Eq. (10).

It is interesting to compare results obtained by Eq. (11) with those reported in Section 3. Despite the flow conditions reported in Fig. (10) are quite different from those used for calibration of Eq. (11), the correlation still shows a reasonable agreement as well as good predictive capabilities once it is applied with exactly the same flow conditions as those of Fig 10. This is shown in Fig. 14. Comparing the results, it can be seen that a good agreement is obtained with the application of the quick Nusselt correlation. Temperature differences are almost always within 20%. It is especially worth noting the very good agreement of the threshold prediction and of the temperature prediction at threshold in this case. The larger discrepancies can be found in the shape of deterioration peak and for the inlet values. The differences are however similar to those obtained in [18] for the comparison with CFD results at lower GD and a different pressure. Overall, the comparison confirms that the correlation allows to get a quick understanding of flow evolution in a wide range of operating conditions.

The discussion of results obtained at different conditions shows that there is an effect of flow Reynolds number in determining the flow evolution under deterioration conditions that, despite it is second order with respect to that of the heat-to-mass flux ratio, cannot be always neglected. The effect of the parameter GD , closely related to Re , on the wall temperature of a methane flow having $p/p_c = 2.0$ is shown in Fig. 15. In this figure, it is easily recognized that such effect is evident only for sufficiently large heat-to-mass flux q_w/G , and especially on the temperature peak. In fact, while for $q_w/G = 100 \text{ J/kg}$ the wall temperature is insensitive to the value of GD , when $q_w/G = 1000 \text{ J/kg}$ the larger the parameter GD the larger the wall temperature.

This result, obtained analyzing the application of the Nusselt correlation to different tube conditions, points out that further studies should be carried out to identify the possible need of correction of the threshold correlation to take into account the role of flow Reynolds number.

5. Asymmetric heating

The analysis performed in the above sections are focused on tubes with circular cross section that are homogeneously heated along their perimeter. This set-up is particularly useful for studying

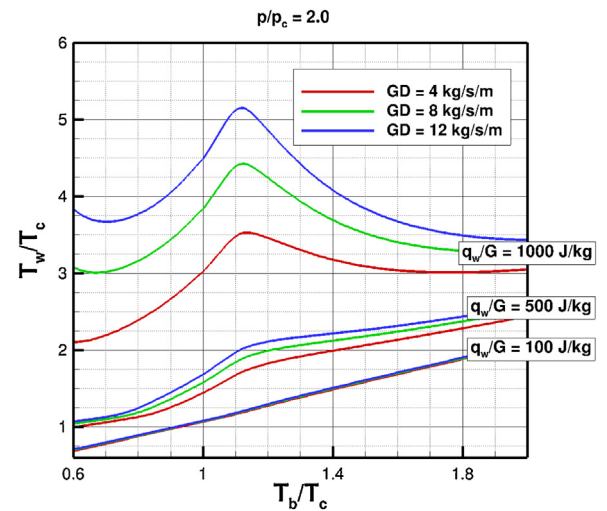


Fig. 15. Reduced wall temperature behavior of a methane flow having $p/p_c = 2.0$.

the basic heat transfer phenomena, because the flow can be easily reduced to one dimensional. However, actual rocket engine cooling channels are oval/rectangular and heating comes across one side only and through a metallic liner that separates the coolant from the combustion gas and metallic ribs that separate the cooling channels. The occurrence of the heat transfer deterioration in actual rectangular cooling channels fed with supercritical methane has been studied in details by the Authors in [27] using the CFD solver developed in [19]. The simplification of having considered a straight channel with constant heat flux at the combustion-gas side, instead of curved channel with variable heat flux as in a typical rocket engine configuration, does not impair the validity of such study, that is, to investigate the possibility to incur in heat transfer deterioration also in case of asymmetric heating. In fact, since channel curvature provides an additional heat transfer mechanism, the inclusion of such effect may shade the basic understanding of the heat transfer deterioration. In particular, the effect of channel curvature is to promote fluid recirculation in the channel cross section. Fluid recirculation in general improves the heat transfer. However, a peculiar behavior is reported in [28] where the analysis focuses on the case of double curvature channels with heating from one side only, which is typical of rocket engine thrust chamber having a convergent-divergent shape. Results show that for a supercritical fluid at pressure much higher than critical the heat transfer can be either enhanced or reduced with respect to the straight case. The possibility of having enhancement or reduction depends on the flow direction with respect to the heating sequence, whether first concave-side and then convex-side heating or vice versa.

The setup presented in [27] is used here to highlight differences and commonalities between asymmetric and uniform heating of coolants from sub-critical temperature and at two supercritical pressures. The geometrical setup (Fig. 16) has similar size as the circular cross section channels discussed above, and also flow Reynolds number is similar, being at inlet about $0.9 \cdot 10^5$. However, in this case also the heat transfer along solid walls has to be considered. Uniform heat flux is enforced on the channel bottom (inner wall) whereas the top of the channel is assumed as adiabatic (outer wall). The present test cases consider methane as the coolant and copper as the solid wall material. Two test cases are considered for this setup having as a common feature the mass flow rate (and thus $GD_h = 10 \text{ kg/m/s}$) and the heat-to-mass flux ratio, which is $q_w/G = 625 \text{ J/kg}$, where the considered heat flux is obviously an averaged value as it changes along the channel perimeter. On the other hand, test cases refer to two different inlet pressures: one closer ($p_i/p_c = 1.27$) and one farther ($p_i/p_c = 2.25$) from

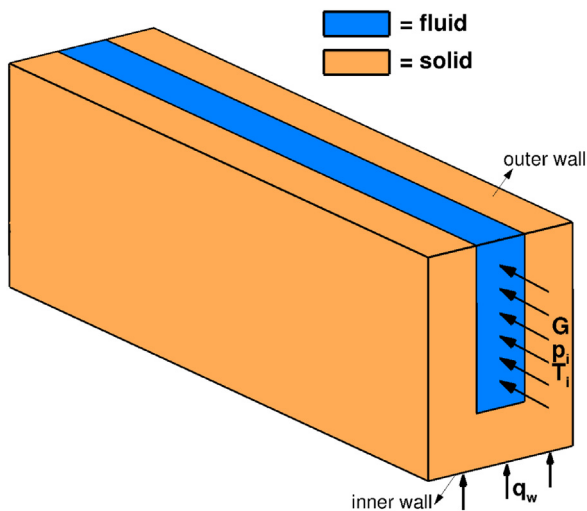


Fig. 16. Schematic of the asymmetric heating test case.

the critical pressure. According to Eq. (10), the heat transfer deterioration threshold for methane is $q_w/G = 277 \text{ J/kg}$ for the case with $p_i/p_c = 1.27$ and $q_w/G = 471 \text{ J/kg}$ for the case with $p_i/p_c = 2.25$. Therefore, deterioration should occur in both cases because of the higher than threshold heat-to-mass flux ratio. Examining the solution obtained at the lower pressure value, it can be seen that deterioration does occur. This is shown by the temperature fields of both coolant and solid material in Fig. 17. The temperature evolution is plotted at different cross-sections, evenly spaced from the inlet toward the exit. Fig. 17 shows that the fluid heats-up as it goes into the channel even if, because of the non-homogeneous heating from the channel perimeter and the limited fluid mixing, an

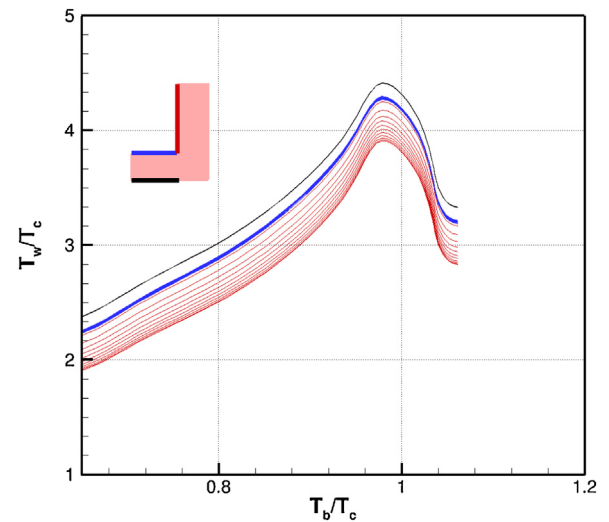


Fig. 18. Wall temperature along the channel at near-critical pressure (reduced inlet pressure is 1.27). (For interpretation of the references to color in the text, the reader is referred to the web version of this article.)

evident methane thermal stratification is present. Fig. 17 also permits to analyze the wall temperature distribution along the channel length, characterized by a peak value at the fourth section. This is a clear mark of the heat transfer deterioration that occurs for the present fluid flow. It is interesting to note that the effect of the heat transfer deterioration is present on the whole cross section, not only at the inner wall. This is highlighted in Fig. 18, that describes the wall temperature evolution along the channel. In particular, the red lines report the value along the internal side of the rib, the blue lines the evolution along the channel base and the black lines the

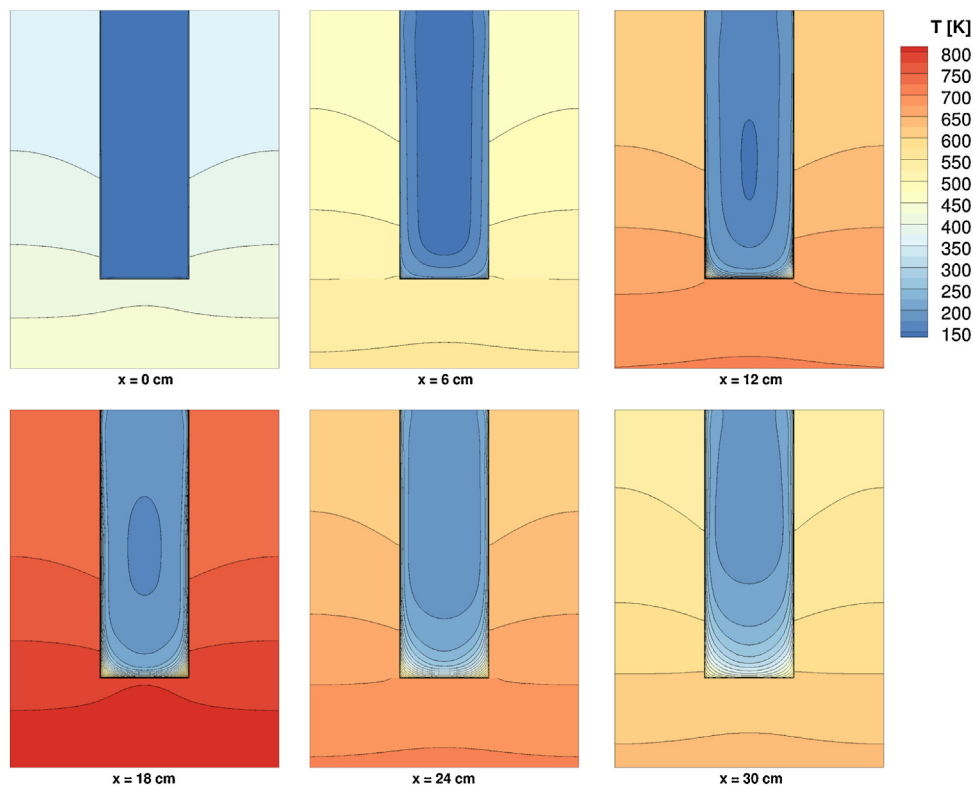


Fig. 17. Temperature fields at different channel sections (reduced inlet pressure is 1.27).

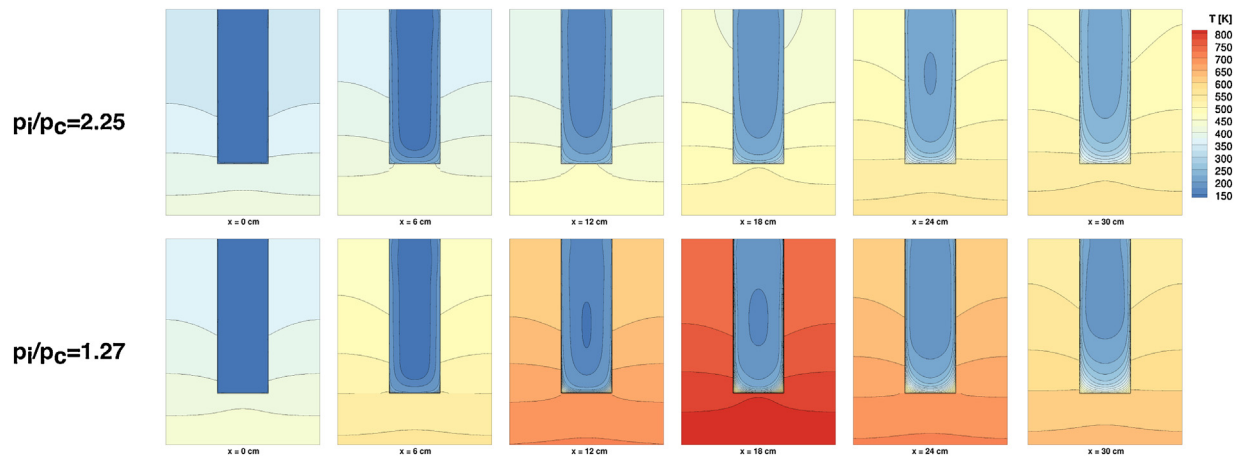


Fig. 19. Comparison of temperature fields at different channel sections at two different flow pressures.

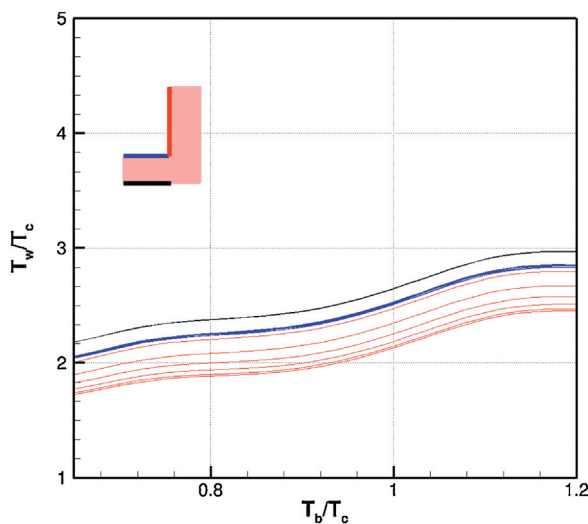


Fig. 20. Wall temperature along the channel at high pressure (reduced inlet pressure is 2.25).

maximum temperature value which occurs along the inner wall. It is evident that the wall temperatures do not change much within each cross section and consequently they evolve as one along the channel. The explanation for this effect is also to be found in the high thermal conductivity of copper. Comparing the wall temperature behavior in the rectangular channel with those discussed in the above sections for circular cross section tubes, it can be observed that they are quite similar to each other despite the asymmetric heating of the rectangular cross section channel.

Different is the case at higher pressure. In this case, in fact, despite the criteria discussed above would foresee heat transfer deterioration, a smooth wall temperature increase is found. This is shown by the comparison of the wall and flow temperature at the different cross section between the cases at the different pressures in Fig. 19. In the figure, the higher pressure evolution is shown on top and is compared with the temperature evolution at lower pressure reported below at the corresponding cross sections. It is clearly seen that no deterioration occurs in this case indicating that the threshold value is expected to be higher in presence of asymmetric heating. It seems therefore that asymmetric heated channels are less prone to present heat transfer deterioration than uniformly heated channels. Also in the high pressure case Fig. 19 and the detailed wall temperature evolution reported in Fig. 20

show that wall temperatures in the cross section evolve as one along the channel.

6. Rough wall channels

One important effect that has been found of major importance in determining the behavior of near critical coolants under heating and the possible occurrence of heat transfer deterioration is that of wall roughness. The role of wall roughness on heat transfer enhancement has been described by suitable correlations in normal conditions, showing that it increases with roughness following the increase of wall friction [29], thanks to the increased turbulence production with surface roughness. However, its increase tends to be less than friction increase up to flatten out in the highest range of roughness.

To evaluate the effect of the wall surface roughness on the heat transfer deterioration, the same flow conditions of the rectangular cross section channel of Section 5 have been simulated for rectangular wall of increasing roughness from smooth channel to relative roughnesses of $\varepsilon/D_h = 0.003$ and $\varepsilon/D_h = 0.0067$, where ε is the absolute equivalent sand grain roughness. The results obtained at the two reduced pressures of $p_i/p_c = 1.27$ and $p_i/p_c = 2.25$ show the expected significant influence of roughness on the evolution of wall temperature along the channel (Fig. 21).

At the higher reduced pressure, as shown in Section 5, no heat transfer deterioration takes place and the increasing roughness yields the expected result of increasing the convective heat transfer and therefore it yields a significant reduction of the wall temperature.

On the other hand, at the lower reduced pressure, heat transfer deterioration occurs in case of smooth walls. Also in this case, the reduction of wall temperature because of the increased convective heat exchange yielded by wall roughness is quite evident from the figure. As a consequence, also heat transfer deterioration evolution is affected by wall roughness. The peak heat flux is reduced and moves towards higher fluid bulk temperatures. At the highest roughness considered the peak barely vanishes, so that heat transfer deterioration no longer occurs. The temperature evolution becomes similar to that seen at higher pressure, with the smooth temperature increase which is anticipated, with respect to the higher pressure case, because of the lower pseudo-critical temperature. In fact, the pseudo-critical temperature is $T/T_c = 1.03$ at $p/p_c = 1.27$, and $T/T_c = 1.14$ at $p/p_c = 2.25$.

These results show that wall roughness has a mitigating effect with respect to heat transfer deterioration especially when high roughness levels are considered, and therefore that criteria devel-

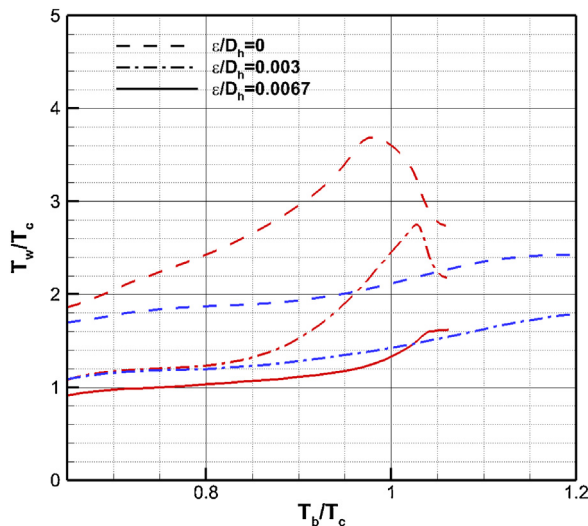


Fig. 21. Comparison of the inner wall temperatures and flow pressure obtained with different channel wall roughness: high pressure (blue lines) and low pressure (red lines). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

oped for smooth surfaces should be considered as tools for a conservative evaluation of the heat transfer deterioration phenomenon. Such mitigating effect is expected also employing heat transfer enhancement structures such as the dimples and ribs because, similarly to the wall roughness, they are able to enhance the flow turbulence. For instance, in [30] it is shown that the heat transfer deterioration occurring in a supercritical pressure methane flow is drastically weakened employing a ribbed cooling channel.

7. Conclusions

An overview of the importance of the possible occurrence of heat transfer deterioration in cooling channels of liquid rocket engines which use light hydrocarbons as the fuels, has been presented. In particular, the achievements of the Authors and their coworkers during the last decade have been summarized. The present review shows that heat transfer deterioration while heating supercritical liquid rocket engine propellants may occur under specific conditions. In particular, it is seen that the high level of heat flux typical of liquid rocket engines can lead to values of heat-to-mass flux ratio that trigger this undesired phenomenon. On the other hand, it has been discussed and highlighted that its possible occurrence is mitigated in case of pressure much higher than critical and by wall roughness. The models for a first evaluation of the required condition for the onset of the heat transfer deterioration and the wall temperature evolution developed for methane and light hydrocarbons have been shown as able to provide conservative predictions. This is because they are based on the case of smooth and uniformly heated tubes which are more prone to deteriorate than the actual rough asymmetrically heated channels. These models pave the way for the possible future refinements able to include the role of asymmetry and wall roughness.

Declaration of Competing Interest

The authors report no declarations of interest.

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